

Problem 1. Given trapezoid $ABCD$ with parallel sides AB and CD , assume that there exist points E on line BC outside segment BC , and F inside segment AD , such that $\angle DAE = \angle CBF$. Denote by I the point of intersection of CD and EF , and by J the point of intersection of AB and EF . Let K be the midpoint of segment EF ; assume that it does not lie on line AB . Prove that I belongs to the circumcircle of ABK if and only if K belongs to the circumcircle of CDJ .

Problem 2. Let O be the circumcentre of an acute-angled triangle ABC with $\angle B < \angle C$. The line AO meets the side BC at D . The circumcentres of $\triangle ABD$ and $\triangle ACD$ are E and F , respectively. Extend the sides BA and CA beyond A , and choose on the respective extensions points G and H such that $AG = AC$ and $AH = AB$. Prove that the quadrilateral $EFGH$ is a rectangle if and only if $\angle ACB - \angle ABC = 60^\circ$.